

Operating Manual for Digital Precision Manometer GDH 11 / GDH 11A

Specification:

Range:	0 to 1100 mbar absolute
Overload:	max. 2 bar abs. (sensor will neither be destroyed nor will a recalibration be necessary)
Resolution:	1 mbar
Accuracy: (unit)	1 mbar +/-1 digit (at nominal temperature of 25° C)
Drift (unit):	GDH 11: 0.05%/K GDH 11A: 0.01%/K
Sensor:	piezoresistive absolute pressure sensor, integrated in measuring device. Sensor suitable for non-corrosive and non-ionising gases and liquids.
Sensor accuracy: (typical values)	+/- 0.2% f.s. (full scale) hysteresis and linearity +/-0.4% f.s. (full scale) temperature influence from 0 to 50° C [optionally: double accuracy: 0.1% resp. 0.2% f.s.]
Working temperature:	0 to 50° C
Display:	3 1/2 digit LCD display, approx. 13 mm high
Power supply:	9V battery type JEC 6 F 22 (included).
Power consumption:	approx. 5 mA
Low battery warning:	"BAT" displayed automatically in case of low battery
Dimensions of case:	impact resistant ABS plastic housing, approx. 150 x 86 x 30 mm (H x W x D) with integrated pop-up clip for table-top or suspended use
Weight:	approx. 230 g (unit ready for operation)

Additionally for GDH 11A

Analog output:	0 - 1V DC equiv. 0 to 1100 mbar, socket for 3.5 mm dia mono plug (plug included in scope of supply)
Mains connection:	2.5 mm socket for external 10 to 12 V DC supply voltage (suitable power pack: GNG 09 or GNG 10) Battery will be automatically disconnected as soon as power pack is connected.

Points to be observed during operation

1. Make sure to apply correct operating voltage as wrong voltage will lead to measuring inaccuracies. As soon as "BAT" is displayed, battery is used up and needs to be replaced.
2. Please avoid temperatures exceeding the permissible range. Measuring device should never be subjected to condensation caused by large temperature differences or quick temperature changes. (Wait for temperature adjustment to ambient temperature to be completed before starting the measuring procedure).
3. The measuring device has to be treated carefully and should only be used in accordance with the aforementioned specifications (do not throw, bump device etc.).
4. Both the unit and the sensor will be calibrated before leaving our works. The sensor being an absolute pressure sensor, the the current atmospheric pressure as is displayed as standard. If necessary adjust potentiometer "Scale" (identification "scale"). **Make sure never to change the zero potentiometer setting (identification: "Nullp."=zero point) !**

If the permissible pressure has been exceeded (overload), recalibration of the unit and sensor may be required. In case you are not in a position to carry out a recalibration yourself, please return **both** sensor **and** unit to our works.

Additionally for GDH 11A:

5. The unit has an analog voltage of 0 to 1 Volt.
The display will show 1100 mbar for an analog voltage of 1.0 Volt.
If necessary, adjust display value to correct value of 1100 using the "scale display" potentiometer (no identification on front plate of unit).